EXECUTIVE SUMMARY

The factors affecting the formation of protective alumina scales on Nb-base alloys by selective oxidation have been extensively investigated. Alumina cannot be formed in air at 1atm. on binary Nb-Al solid solutions or any Nb-aluminides including $NbAl_3$. The existing theoretical knowledge of selective oxidation has been applied to oxidation of Nb-Al alloys and aluminides to determine approaches by which this behavior may be altered. The effects of Al-content, temperature, atmosphere, third element additions, and microstructure on the transition from internal to external oxidation of aluminum have been evaluated and the the conditions under which protective alumina scales can be formed on Nb-Al alloys has been defined.

Third element additions are required to form alumina by selective oxidation of Al on Nb-base alloys. The most effective additions are those which can reduce the solubility and diffusivity of oxygen in the alloy, enhance the diffusion of Al in the alloy, and limit transient oxidation. The addition of high e/a elements such as Mn and Ru, which should reduce solubility and diffusivity of oxygen and suppress the internal oxidation of Al, were not effective. These additions produced transient oxides that grew with linear kinetics at very rapid rates. The addition of oxygen active elements such as Zr, Hf, and Be, which should reduce oxygen diffusivity, also was not effective. Ternary alloys with these additions also oxidized with fast linear kinetics. Titanium was added to alter alloy microstructure and was most effective of all third element additions. Large additions of Ti produced a high temperature B2 phase in which diffusivity of Al is rapid and on which protective alumina scales could be formed.

Although alumina scales could be formed on Nb-Ti-Al alloys with 44at.% Al in air at 1400°C, the rates of oxidation were 1-2 orders of magnitude greater than those of other alumina formers such as NiAl. This was the result of a slow growth rate of alumina coupled with a high growth rate of the base alloy transient oxide. It was established that Nb-Ti-Al alloys must be modified with a fourth element in order to form transient oxides which grow with slow linear or parabolic kinetics. Additions of Cr, V, and Si were identified as most promising for reducing transient oxidation and enhancing alumina formation on Nb-Ti-Al alloys.

Additions of Cr and V to Nb-Ti-Al alloys were very effective in promoting formation of alumina at temperatures above 1200°C The appeared to have little effect on transient oxides and are believed to function by reducing solubility and diffusivity of oxygen in the alloy. A synergism was found in which these two elements together were more effective than either one alone. Additions of 3 at.% Cr and 4 at.% V resulted in protective alumina formation at 32 at.% Al in air at 1400°C. Similar results were obtained by additions of Mo in place of V. The alloys did not form alumina at lower temperatures as a result of fast kinetics for transient oxidation. The best balance of composition in at.% is 24Nb, 24Ti, 44Al, 3Cr, 5V for this class of alloys.

The addition of 3-10% Si to Nb-Ti-Al alloys was found to reduce the amount of transient oxidation. This reduced the temperature at which continuous alumina scales could be formed. Best performance was found for a Nb-Ti-Al-Cr-V-Si alloy with 10 at.% Si and 32 at.% Al which formed protective alumina scales in air at 1100°C. Alumina forming behavior was improved by substituting 10% Si for Al in Nb-Ti-Al-Cr-V alloys, however, the rates of oxidation were still 2 orders of magnitude greater than that of NiAl at 1100°C. Nb-Ti-Al alloys modified with Cr+V and Cr+V+Si to form alumina scales without significant transient oxidation oxidized with slow parabolic kinetics at the same rate as NiAl from 1300-1600°C.

The oxidation behavior of Nb-Ti-Al-Cr-V alloys was studied in greater detail to establish the effect of significant composition and microstructural variables on alumina formation. The major effect of Al content of the alloy was found to be on the temperature at which a transition from internal to external oxidation of Al will occur and the amount of transient oxidation that occurred before it was cut off by formation of a continuous alumina scale. The transition temperature and the amount of transient oxidation decreased as Al content increased. There was no effect of Al content on oxidation kinetics of alloys that did not form continuous alumina scales.

The ratio of Nb:Ti in the alloy also was an important variable. The amount of internal oxidation of Al was reduced and the transition from internal to external oxidation of Al occurred at lower Al contents when the Nb:Ti ratio was increased. The best ratio is in the range of 1-1.3. At ratios higher than 1.3, the alloys are not alumina formers. At lower ratios, the alloys are

γ-TiAl modified with Nb. Alloys in the range of 1-1.3 are single phase B2 above 1400°C and have a two phase γ-TiAl + Nb_2Al (sigma) structure at lower temperature

Rapid solidification processing was evaluated as an approach to improving oxidation resistance of Nb-Ti-Al-Cr-V alloys. Best results were obtained by undercooling below 125°C to retain a stable B2 structure in the alloy at room temperature. On heating, however, the alloy rapidly transformed to the basic two phase microstructure. The finer grain size that was produced by RSP enhanced alumina diffusion, resulting in formation of alumina scales at temperatures below 1400°C. However, the effect was minimal with respect to reducing the temperature for alumina formation and RSP was effective only at temperatures close to the transition temperature. It was effective only if the alloy was a borderline alumina former.

Surface enrichment with Al and Si by ion implantation also was investigated as an approach to enhancing alumina formation. Results were similar to those achieved by rapid solidification processing. The transition temperature for alumina formation was reduced slightly but no effect on kinetics below the transition temperature was observed. Ion implantation was effective only under conditions where the alloy was a marginal alumina former.

Preoxidation was evaluated as an approach to improving oxidation resistance at low temperature where alumina scales could not be formed. Alumina scales preformed in air or oxygen at 1500°C remained protective at 1100°C provided that the sample was not cooled to room temperature after preoxidation. The thermal stresses generated in one thermal cycle were sufficient to crack the preformed alumina scales and no protection was provided on re-exposing to the lower temperature range.

Niobium-Al alloys modified to form alumina must contain at least 32 At.% Al and are intermetallic compounds (aluminides). Results of the investigation indicate little possibility to form protective alumina scales on BCC Nb-Al solid solution-type alloys with a lower concentration of Al. The major factor that precludes alumina formation in such alloys is the low solubility of Al in Nb coupled with the high solubility and diffusivity of oxygen in the BCC solid solution. In addition, the base alloy oxides grow with fast linear kinetics

which precludes alumina formation. The only solution to this problem appears to be the addition of large amounts of Ti to stabilize a high temperature B2 phase on which alumina can be formed. The type of alloying elements and the concentrations necessary to ensure alumina formation lowers the alloy solidus temperatures to around 1650°C. Additions of Si further reduced the solidus temperature to below 1600°C.

Publications

1. R.A. Perkins, K.T. Chiang, and G.H. Meier, "Formation of Al on Nb-Al Alloys..", Scripta Met., v22, p419, 1988.
2. R.A. Perkins, K.T. Chiang, G.H. Meier, and R. Miller, "Formation of Alumina on Nb and Ti-Base Alloys", in Oxidation of High Temperature Intermetallics, T. Grobstein and J. Doychak, eds., The Minerals, metals, and Materials Society, Warrendale, Pa., p157, 1989.
3. G.H. Meier, "Fundamentals of the Oxidation of High Temperature Intermetallics", IBIB, p1.
4. R.A. Perkins and G.H. Meier, "Oxidation Resistant Aluminides for Metal Matrix Composites", to appear in Proceedings of Advanced Materials Conference, Denver, CO., March, 1989.
5. R.A. Miller, G.H. Meier, and R.A. Perkins, "A Study of the Effects of Alloying and Rapid Solidification on the Oxidation of Nb-Ti-Al Alloys", manuscript in preparation for submission to Oxidation of Metals.
6. G.H. Meier and R.A. Perkins, "Effect of Ion Implantation on the Selective Oxidation of Aluminum from Nb-Base Alloys", manuscript in preparation for submission to Scripta Met.
7. G.H. Meier, R.A. Miller and R.A. Perkins, "Effect of Preoxidation on the Oxidation Behavior of Nb-Ti-Al-Cr-V Alloys", manuscript in preparation for submission to High Temperature Technology.

Professional Personnel

The following professional personnel have participated in the program:

LMSC

Mr. R.A. Perkins

Dr. K.T. Chiang

Dr. J.S. Lee